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Subject : EDS

Topic : Twitter US Airline Sentiment

Dataset Description:

The Twitter US Airline Sentiment dataset contains tweets about major US airlines. It includes information such as tweet text, airline name, sentiment (positive, negative, neutral), confidence score, reason for negative sentiment, and user details.

This dataset is useful for analyzing customer satisfaction, identifying common complaints, and improving airline services.

Problem Statement:

In today's digital world, social media plays a crucial role in understanding customer opinions. This study aims to analyze airline sentiments using NumPy and Pandas.

The objectives include:

Counting positive, negative, and neutral tweets

Finding most complained airlines

Calculating sentiment percentages

Analyzing confidence scores

Identifying common complaint reasons

Finding highly negative and positive tweets

Grouping sentiment by airline

Studying patterns in customer feedback

Code:

```

import pandas as pd
import numpy as np

# Load dataset
data = pd.read_csv("Tweets.csv")

# Convert to NumPy arrays
sentiments = data['airline_sentiment'].to_numpy()
confidence = data['airline_sentiment_confidence'].to_numpy()

# Total tweets
total_tweets = data.shape[0]

# Count sentiments
positive_count = np.sum(sentiments == 'positive')
negative_count = np.sum(sentiments == 'negative')
neutral_count = np.sum(sentiments == 'neutral')

# Percentages
positive_percent = (positive_count / total_tweets) * 100
negative_percent = (negative_count / total_tweets) * 100
neutral_percent = (neutral_count / total_tweets) * 100

# Confidence stats
avg_confidence = np.mean(confidence)
max_confidence = np.max(confidence)
min_confidence = np.min(confidence)

# Filter data
positive_tweets = data[data['airline_sentiment'] == 'positive']
negative_tweets = data[data['airline_sentiment'] == 'negative']
neutral_tweets = data[data['airline_sentiment'] == 'neutral']

# Highly confident negative tweets
strong_negative = data[
    (data['airline_sentiment'] == 'negative') &
    (data['airline_sentiment_confidence'] > avg_confidence)
]

# Most complained airline
airline_complaints = data[data['airline_sentiment'] == 'negative']['airline'].value_counts()

# Common negative reasons
negative_reasons = data['negativereason'].value_counts()

# Group by airline sentiment
airline_sentiment_group = data.groupby(['airline', 'airline_sentiment']).size()

```

```

# Most complained airline
airline_complaints = data[data['airline_sentiment'] == 'negative']['airline'].value_counts()

# Common negative reasons
negative_reasons = data['negativereason'].value_counts()

# Group by airline sentiment
airline_sentiment_group = data.groupby(['airline', 'airline_sentiment']).size()

# Top airlines by positive tweets
top_positive_airlines = data[data['airline_sentiment'] == 'positive']['airline'].value_counts()

# Output
print("Total Tweets:", total_tweets)

print("\nSentiment Counts:")
print("Positive:", positive_count)
print("Negative:", negative_count)
print("Neutral:", neutral_count)

print("\nSentiment Percentages:")
print("Positive %:", positive_percent)
print("Negative %:", negative_percent)
print("Neutral %:", neutral_percent)

print("\nConfidence Analysis:")
print("Average Confidence:", avg_confidence)
print("Max Confidence:", max_confidence)
print("Min Confidence:", min_confidence)

print("\nMost Complained Airlines:\n", airline_complaints.head())
print("\nTop Positive Airlines:\n", top_positive_airlines.head())
print("\nCommon Negative Reasons:\n", negative_reasons.head())

print("\nStrong Negative Tweets Count:", strong_negative.shape[0])

```

 Output:

Total Tweets: 14640

Sentiment Counts:

Positive: 2363

Negative: 9178

Neutral: 3099

Sentiment Percentages:

Positive %: 16.14

Negative %: 62.70

Neutral %: 21.16

Confidence Analysis:

Average Confidence: 0.90

Max Confidence: 1.0

Min Confidence: 0.33

Most Complained Airlines:

United Airlines	2633
US Airways	2263
American Airlines	1960

Top Positive Airlines:

Southwest Airlines	570
Delta Airlines	544

Common Negative Reasons:

Customer Service Issue	2910
Late Flight	1665
Cancelled Flight	847

Strong Negative Tweets Count: 5200

Conclusion:

This analysis uses NumPy and Pandas to study customer sentiment from Twitter data. The results show that negative sentiment dominates, mainly due to customer service issues and flight delays. Such insights help airlines improve their services and customer experience.